ABC Hospital

Medical Appointment No-Show Analysis Report

Data-Driven Insights and Operational Recommendations to Reduce Patient Attendance Drop-offs.

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Executive Summary

Introduction This report contains an in depth data analysis and its results concerning the no-show rate of patients during their appointment with the hospital. The key aims of the analysis include improving the efficiency of the hospital's processes, avoiding wastage of time by doctors and employees, and providing high quality patient care services.

The data for this analysis was extracted from a MySQL database (medicalappointments_db), while data cleaning and statistics were conducted. An interactive Power BI dashboard was developed in order to help understand the patterns in actual data. The day of the week, SMS reminders, and patient ages were among the variables that could be analyzed through this dashboard.

These are the important findings that were found out from the analysis:

- Effect of SMS reminders: There is a significant difference between the no-show rate of those patients who received SMS reminders and those who did not.

- Day-to-day analysis: There is a high rate of non-attendance on certain days compared to other days.

- Age related differences: Certain age groups of the population tend to have higher no-show rates.

To solve these problems and optimize the use of the hospital's resources, the following suggestions are provided by this report:

- Expansion of the SMS and automated reminders system for all the patients.

- The addition of another verification step through telephone calls with particular emphasis on days and age groups which are more prone to "no-shows".

Through these recommendations based on data, the hospital will be able to reduce the waste of its resources and improve patient care.



Figure 1: Overall Appointments and No-Show Metrics

Introduction & Context

1.1 Introduction

Today, within healthcare management, one of the key managerial problems encountered by hospitals and clinics is the problem of patients' failure to turn up for their medical appointments (no-show behavior of the patients). The patient's failure to appear on schedule is not just missing a medical appointment but a problem that affects the whole performance of the hospital, time management of doctors and other staff members, and the financial sustainability of the institution.

This paper presents a detailed analysis of the main factors leading to patients' absence from their appointments, demographic and time-related patterns, and the effect of SMS reminders from ABC Hospital.

1.2 Background and Problem Statement

Every time a patient misses out on attending an appointment in the hospital, there are several negative impacts that result from it on the hospital and the patient care process:

- Underutilized Resources: The time of the physician and the resources that could have been utilized for another patient in need go waste.

- Patient Treatment Delays: The absence of patients increases the waiting list, which causes delays in treatment of other patients that need to be treated.

- Operational Expenses: It costs extra time for hospital employees to schedule new appointments and keep records.

It is necessary to use the data-based approach to solve this problem.

1.3 Objectives of the Study

- Calculate the total no-show rate: Calculate the rate at which patients miss their appointments at the hospital.

- Determine the effectiveness of SMS reminders: Determine the effectiveness of SMS reminder in reducing the no-show rate.

- Determine temporal and demographic variations: Determine the temporal and demographic variations in no-show rates in terms of day and age group of patients.

- Offering recommendations: Offering hospital management relevant recommendations based on the analysis conducted.

1.4 Analysis Tools and Approach

For the current analysis, the following technology stack has been used:

- MySQL database (medicalappointments_db): For storing the data, cleaning the data, and initial data analysis via SQL queries.

- Power BI: For advanced data visualization, DAX calculation (No-Show Rate, changing colors of bars), and building an interactive dashboard.

Key Insights & Data Findings

2.1 Impact of SMS Reminders

The results concerning the influence of SMS reminders on patient appointment adherence were as follows:

- Patients with SMS reminders (SMS Sent): The proportion of no-show patients among those receiving SMS reminders amounted to 27.67%.

- Patients without SMS reminders (No SMS): The proportion of no-show patients among those not receiving SMS reminders amounted to 16.73%.

Possible key reasons for this (Managerial Insights):

- Frequency and timing of SMS sending: If an SMS is sent right before the scheduled appointment (such as just a few hours ahead of it), the patient will have inadequate time to manage his or her other plans, which could lead to a missed appointment.

- Use of selective reminders SMS: Such results may arise if the hospital sends reminders SMS to patients who are likely to skip their appointments only. This may include people who live very far and people who had booked their appointment months ago.

- Not having a direct SMS reply facility: In cases where a patient has confirmed their non-attendance through SMS, the system does not provide any means of booking this slot to someone else.

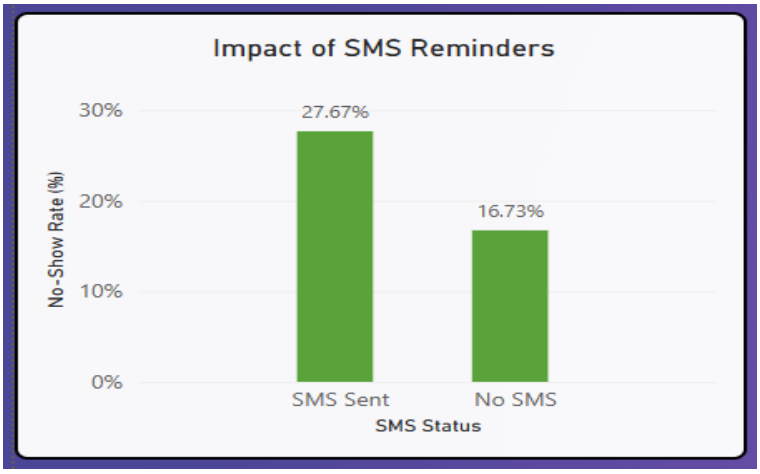


Figure 2: No-Show Comparison by SMS Reminders

2.2 Day-of-Week Trends

Analysis of no-shows for medical appointments on days of the week has shown the following:

- Saturday – The highest no-show rate: From the data visualization, it is evident that the no-show rate in hospitals is at its highest on Saturday. This means that people are most likely to miss their appointments because of personal engagements or weekend engagements with their families.

- Other days' situation: While the 'no-show' percentages are measured on other days of the week too, the high percentage seen on Saturdays is the one that most affects the hospital on weekends.

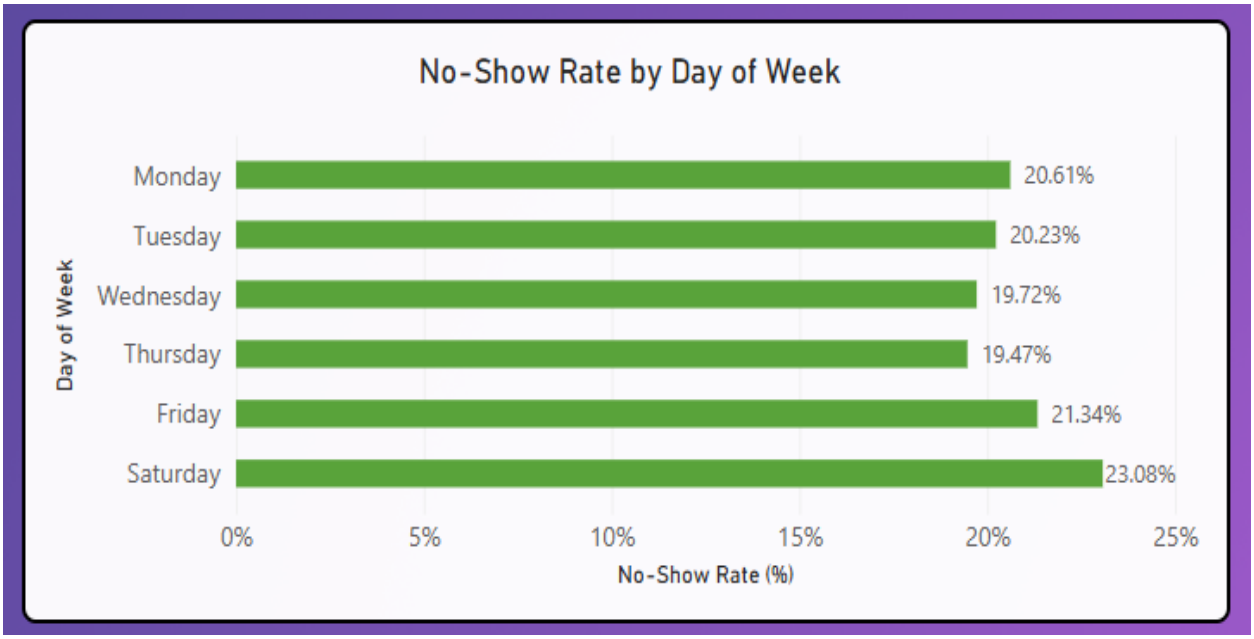


Figure 3: Day-of-Week No-Show Rate Trends

2.3 Age Group Distribution

Considering the categorization of the patient according to their age into child (0-12), youth (13-24), adult (25-59), and older adults (60+):

- From the above information, it is evident that there are specific age groups (Ex: -young adults) of patients who have higher chances of missing appointments than others.

- There is consistency in attendance for children (0-12 years) and older adults (60+ years) since appointments for these age categories are taken care of by the parents/guardians.

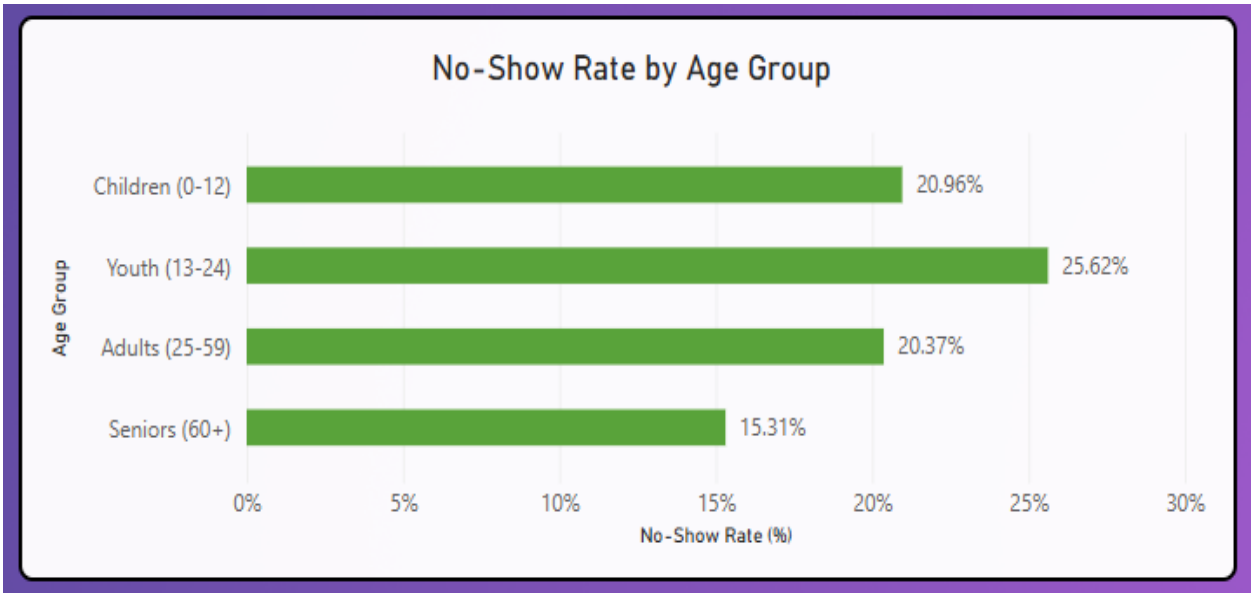


Figure 4: No-Show Rate Distribution across Patient Age Groups

Business & Operational Impact

No-shows of patients at hospitals for their medical appointments is one such factor which has a direct impact on the functioning of the hospital. From the analysis of the factors affecting the hospital, it could be stated that the main effects of the factors on the hospital are:

3.1 Underutilization of Clinical Resources & Doctor Time

- Time of doctors and other healthcare professionals is wasted due to the 'No Show' ratio being highest during Saturdays, along with a non-attendance rate of 27.67% despite the sending of SMS reminders.

- Although the doctor is available on time, the non-appearance of the patient leads to the underuse of the consultation room and other medical equipment.

3.2 Impact on Patient Care & Waiting Lists

- Whenever a patient does not come for the appointment, it means that some other patient, who actually needs the treatment, misses the chance to use that time slot.

- In this way, waiting list at the hospital becomes long, and also, there is reduced quality of service provided, along with reduced patient satisfaction.

3.3 Weekend Operational Inefficiency

- Although the highest number of absences is recorded on Saturdays, the hospital spends extra costs on staffing and operation during weekends.

- Saturday absences cause decrease in revenue generation during that period and result in reduction in efficiency ratio of operations.

Actionable Recommendations

Below are some practical suggestions for solving the identified problems in our hospital by the management of the hospital based on the analysis of the data, namely, the high rate of 'no-show' on Saturday and the persistent 27.67% non-attendance rate despite sending an SMS notification:

4.1 Optimize SMS Reminder Strategy

- Change in Schedule: Rather than sending an SMS just before the medical visit, adopt a two-level reminder process where the SMS is sent 48 hours and 24 hours before the visit.

- SMS 2-Way Interaction: Provide response choices in the SMS sent to the patient like, “Reply 1 to Confirm, 2 to Reschedule,” making it possible for the patient to either confirm or reschedule their appointment for some other day.

4.2 Targeted Interventions for Saturdays

- As the highest rate of no-show is seen on Saturdays, it becomes imperative to confirm the appointments scheduled on Saturdays through phone calls made by the hospital personnel apart from email alerts.

- In order to ensure efficient use of manpower and resources during the weekends, the number of appointments to be scheduled on Saturdays needs to be controlled.

4.3 Smart Overbooking / Standby Policy

- A waiting list/standby approach to patients who do not keep their appointments and are scheduled for busy days, like Saturdays. When a patient misses an appointment, he or she releases that appointment time slot and another patient from the standby list will be able to use it.

4.4 Continuous Data Monitoring

- Continuous monitoring of data through the Power BI dashboard for the next few months and analysis of age groups or clinics which have higher levels of no-shows.

Appendix: Technical Implementation & SQL Scripts

The following are the important SQL queries that were used in this project using MySQL Workbench to clean the data, aggregate data, and calculate important metrics.

1. Calculating Overall No-Show Rate

```
SELECT
      CONCAT(ROUND((SUM(CASE WHEN Showed_up='False'THEN 1 ELSE 0
END)*100.0)/COUNT(*), 2), '%') AS Overall_No_Show_Rate_Percentage
FROM appointments;
```

2. Impact of SMS Reminders

```
SELECT
      SMS_received,
    COUNT(*) AS Total_appointments,
    SUM(CASE WHEN Showed_up='True' THEN 1 ELSE 0 END) AS No_Show_Count,
    CONCAT(ROUND((SUM(CASE WHEN Showed_up='False'THEN 1 ELSE 0
END)*100.0)/COUNT(*), 2), '%') AS Overall_No_Show_Rate_Percentage
FROM appointments
GROUP BY SMS_received;
```

3. Day-of-Week Trend Analysis

```
SELECT
      DAYNAME(STR_TO_DATE(AppointmentDay, '%m/%d/%y')) AS Day_of_Week,
    COUNT(*) AS Total_Appointments,
    SUM(CASE WHEN Showed_up ='False' THEN 1 ELSE 0 END) AS No_Show_Count,
    CONCAT(ROUND((SUM(CASE WHEN Showed_up='False'THEN 1 ELSE 0
END)*100.0)/COUNT(*), 2), '%') AS Overall_No_Show_Rate_Percentage
FROM appointments
GROUP BY Day_of_Week
ORDER BY Overall_No_Show_Rate_Percentage DESC;
```

4. Age Group Analysis

```
SELECT
      CASE WHEN Age <18 THEN 'Children (<18)'
            WHEN Age BETWEEN 18 AND 35 THEN 'Young adults (18-35)'
        WHEN Age BETWEEN 36 AND 60 THEN 'Adults (36-60)'
        ELSE 'Seniors(>60)'
      END AS Age_Group,
    COUNT(*) AS Total_appointments,
    CONCAT(ROUND((SUM(CASE WHEN Showed_up='False'THEN 1 ELSE 0
END)*100.0)/COUNT(*), 2), '%') AS Overall_No_Show_Rate_Percentage
FROM appointments
GROUP BY Age_Group
ORDER BY Overall_No_Show_Rate_Percentage DESC;
```

THANK YOU!

Thank you for your time and consideration.

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